Buying New, Buying Second-Hand, or Renting?

A Life-Cycle and AHP-Based Decision Analysis of Formalwear for Students

Yang Yihua, Cao Xiaoyue, Yang Liu, Liu Ziqing, and Xie Tongling

Abstract

Students choosing whether to purchase or rent formalwear must balance carbon emissions, expenditure, and convenience. This study compares purchasing a new dress, purchasing a second-hand dress, and renting the same dress model for a National University of Singapore student attending six formal events over one year. Life cycle assessment (LCA) estimates allocated annual greenhouse gas emissions, while life cycle costing (LCC) quantifies direct consumer expenditure. These outcomes and a time-based convenience indicator are combined through analytic hierarchy process (AHP)-weighted composite scoring. Under baseline assumptions, annual emissions are 4.92 kg CO_2e for each purchase option and 11.38 kg CO_2e for rental. Annual expenditure is SGD 60.99, 72.00, and 108.00 for second-hand purchasing, new purchasing, and rental, respectively. Second-hand purchasing ranks first, reflecting its lower cost and identical modelled emissions and active time relative to new purchasing. Given rental's substantial collection-and-return burden, transport sensitivity analysis tests how travel attribution affects the ranking. The results support second-hand purchasing in the baseline scenario and identify substantial reductions in attributed travel burdens as a condition for rental competitiveness.

Keywords: formalwear; life-cycle assessment; life-cycle costing; analytic hierarchy process; transport sensitivity

1 Introduction

Formalwear acquisition combines garment production, care, payment, and personal effort. Purchasing enables repeated use, while rental provides temporary access supported by collection, return, and cleaning services. Their relative performance depends on garment utilization and the operations required to serve each occasion. Clothing inventories illustrate the importance of production and use-phase assumptions (Sandin et al., 2019); research on formalwear rental further identifies use frequency and service arrangements as determinants of environmental performance (Monticelli & Costamagna, 2023).

This study compares three acquisition strategies for the Love, Bonito Eyka Multi-way Satin Column Dress, size S: buying new, buying second-hand, and renting for each event. A common annual scenario links carbon accounting and consumer expenditure to AHP-weighted evaluation. Transport sensitivity analysis then examines how collection-and-return arrangements influence the preferred option (Figure 1).

2 Methods

2.1 Scenario, evidence, and assumptions

The functional unit is the provision of six formalwear uses for one NUS student over one year. Each purchase option involves one acquisition and six wears; rental involves six separate seven-day bookings. The scenario permits repeated use of the same style and assumes suitable fit, dress-code compliance, hygiene, and availability. Figure 2 defines the common product system, its resource inputs, and its service and emission outputs. Manufacturing, distribution and retail, use and care, and eventual disposal follow the life-cycle grouping in formalwear LCA (Monticelli & Costamagna, 2023). Annual activities comprise cleaning and acquisition-related delivery or passenger travel. Packaging is reused, and the analysis focuses on the dress and its acquisition service.

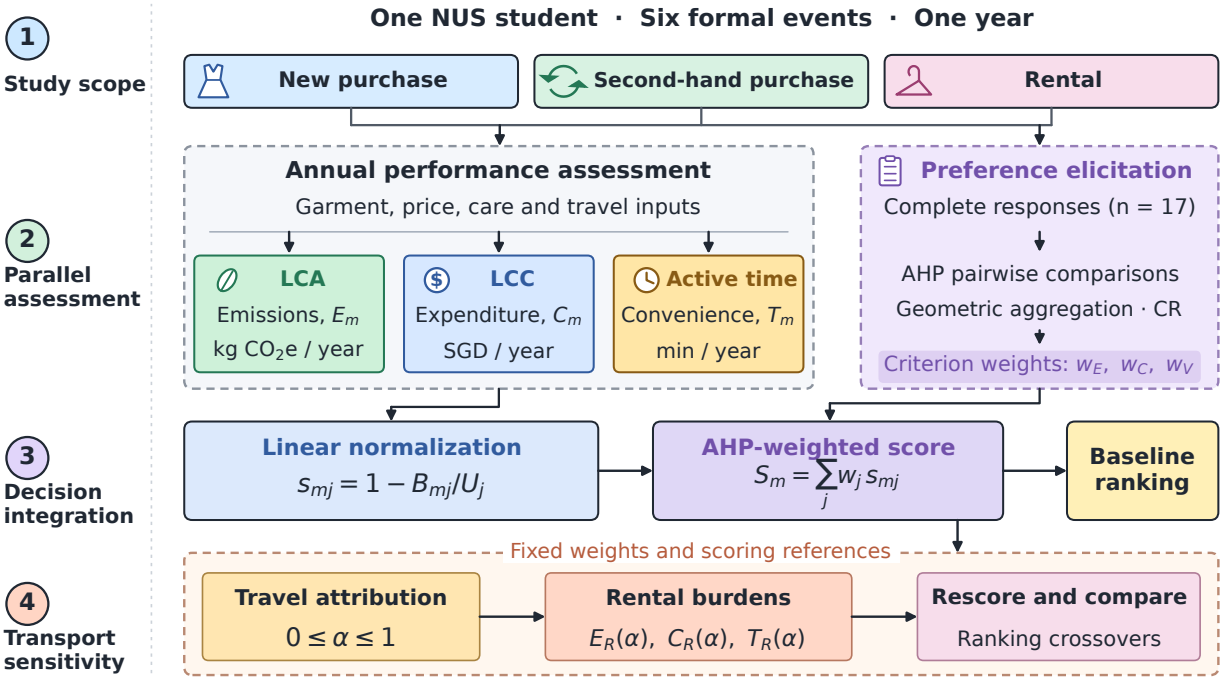


Figure 1: Research framework linking annual performance assessment and AHP weighting to baseline ranking and transport sensitivity analysis.

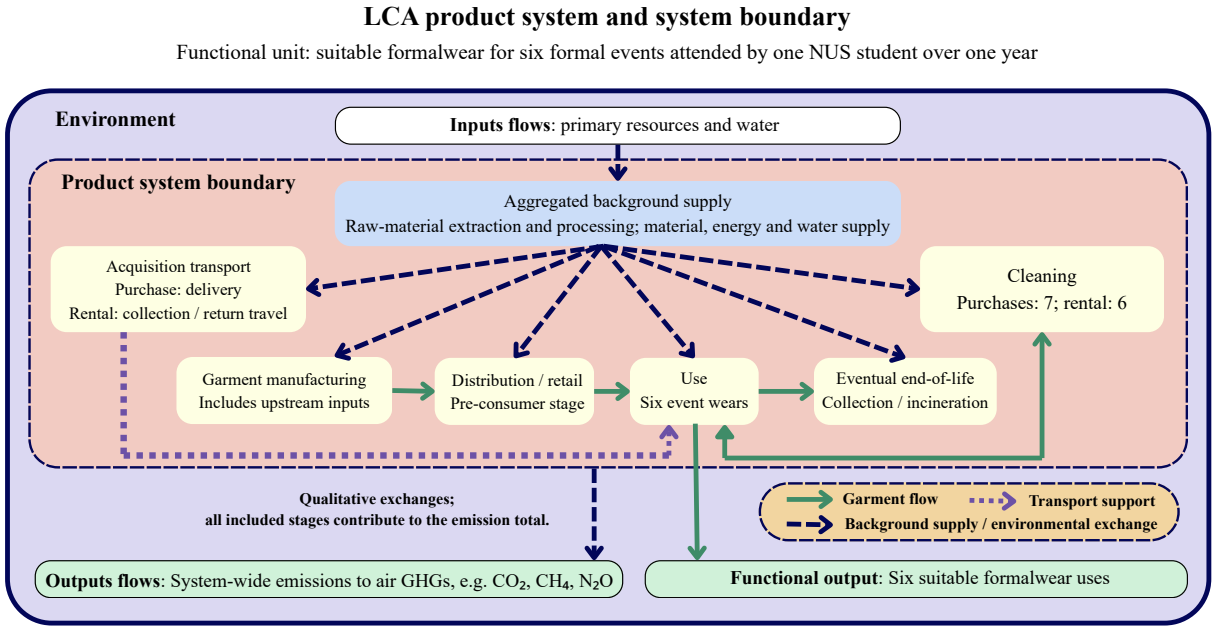


Figure 2: Product system and system boundary for the common functional unit. Resource and emission arrows represent qualitative flows; background supply is embedded in the stage factors. Climate-change impacts are quantified using aggregated literature factors and scenario-specific annual activities.

Advertised prices and terms were archived on 3 November 2026 from the manufacturer and two Singapore listings (forest_green66, n.d.; igotgr8deals, n.d.; Love, Bonito, n.d.). Garment-stage factors are rounded graphical extractions from Sandin et al. (2019), Figure 4.24. The cleaning proxy represents residential washing, drying, and ironing in an Italian study (Monticelli & Costamagna, 2023), Table 4; UK transport factors include direct and upstream energy emissions (Department for Energy Security and Net Zero, 2026). These proxies support a screening assessment.

Table 1 summarizes the principal inputs. All channels share a 26-wear lifetime, and the second-hand garment's reported prior wear satisfies the remaining-capacity condition $1 + 6 \leq 26$.

Table 1: Principal inputs for the six-event annual scenario.

Input	Value	Basis
Purchase price (new / second-hand)	SGD 65 / 52	Advertised
Delivery fee (new / second-hand)	SGD 0 / 1.99	Advertised
Rental fee (seven-day booking)	SGD 10	Advertised
Refundable rental deposit	SGD 99	Advertised
Manufacturing / distribution / disposal	14.8 / 0.84 / 0.99 kg CO_2e/garment	Literature
Cleaning emissions	0.109 kg CO_2e/clean	Literature
Rail emissions	0.0287 kg CO_2e/passenger-km	UK proxy
Van emissions	0.31844 kg CO_2e/vehicle-km	UK proxy
Annual uses / lifetime uses	6 / 26 wears	Scenario
Annual cleans (purchase / rental)	7 / 6	Scenario
Purchase delivery allocation	Two 10-km legs / 20 items	Scenario
Rental collection and return	24 legs × 10 km	Scenario
Home-care cost / active time	SGD 1 / 15 min per clean	Scenario
Rail fare / travel time	SGD 2 / 30 min per leg	Scenario
Purchase ordering time	15 min per order	Scenario
Rental booking and handover time	10 min per booking	Scenario

2.2 Annual emissions, expenditure, and time

Let $m \in \{N, S, R\}$ denote new, second-hand, and rental. With $n = 6$ annual wears, $L = 26$ lifetime wears, and per-garment manufacturing, distribution, and gross disposal factors M, D, F, annual emissions are

$$E_m = \frac{n}{L}(M + D + F) + k_m e_c + d_m e_v + p_m e_r, \tag{1}$$

where k_m denotes cleaning count, d_m allocated delivery vehicle-kilometres, p_m passenger-kilometres, and e_c, e_v, e_r the corresponding emission factors. The fraction $n/L = 6/26$ assigns the share of garment-stage impacts associated with six of the garment's 26 lifetime wears to the annual functional unit. This service-based allocation applies equally across channels, with avoided-production and energy-recovery credits set to zero. The distribution factor D represents the garment's distribution and retail stage; d_m and p_m represent acquisition-related transport in the annual scenario.

Purchase options include one initial clean and one clean after each wear, giving $k_N = k_S = 1+6 = 7$; rental includes one clean after each return, giving $k_R = 6$. The delivery scenario assumes two 10-km legs shared equally by 20 garment-equivalent items, giving $d_N = d_S = (10 \times 2)/20 = 1$ allocated vehicle-km per purchase. Rental requires a collection round trip and a return round trip per booking, yielding $p_R = 6 \times 4 \times 10 = 240$ passenger-km. Thus $(d_N, d_S, d_R) = (1, 1, 0)$ and $(p_N, p_S, p_R) = (0, 0, 240)$.

Defining the common annual garment-stage allocation as G, substitution into Eq. 1 gives the following emissions in kg CO_2e/year:

$$\begin{aligned} G &= \frac{6}{26}(14.8 + 0.84 + 0.99) \approx 3.83769, \\ E_N = E_S &= G + 7(0.109) + 1(0.31844) \approx 4.92, \\ E_R &= G + 6(0.109) + 240(0.0287) \approx 11.38. \end{aligned} \tag{2}$$

The LCC calculation measures direct annual consumer expenditure. Purchase payments enter in full, with resale receipts and residual values set to zero:

$$\begin{aligned} C_N &= 65 + 7(1) = 72.00, \\ C_S &= 52 + 1.99 + 7(1) = 60.99, \\ C_R &= 6(10) + 24(2) = 108.00. \end{aligned} \tag{3}$$

These totals are expressed in SGD/year. Purchase care covers seven home cleans, while rental travel covers 24 one-way fares. Rental care surcharges and additional platform charges are set to zero. The fully refundable SGD 99 deposit represents a temporary liquidity requirement. Carbon allocation and consumer expenditure thus reflect annual service use and annual cash payments, respectively.

Convenience is represented by active consumer time, with shorter durations indicating greater convenience. Home-care time covers active handling, and rental cleaning is assigned to the service provider in the scenario. Ordering and home care yield $T_N = T_S = 15 + 7(15) = 120$ minutes/year. Booking, handover, and dedicated travel yield $T_R = 6[10 + 4(30)] = 780$ minutes/year. These time inputs are scenario assumptions.

2.3 AHP weights and composite scoring

The original questionnaire contains 18 records, of which 17 provide complete comparisons of emissions (E), cost (C), and convenience (V). Q8, Q9, and Q10 supply the ratios E/C, E/V, and C/V.

Each matrix entry $a_{ij}^{(r)}$ expresses respondent r's assessed importance of criterion i relative to j, with $a_{ji}^{(r)} = 1/a_{ij}^{(r)}$ and $a_{ii}^{(r)} = 1$. Reciprocal matrices are aggregated geometrically with equal respondent weights (Forman & Peniwati, 1998). The group priorities are normalized row geometric means, which equal the principal eigenvector for a positive reciprocal 3×3 matrix:

$$\bar{a}_{ij} = \left(\prod_{r=1}^{17} a_{ij}^{(r)} \right)^{1/17}, \qquad g_i = \left(\prod_{j=1}^{3} \bar{a}_{ij} \right)^{1/3}, \qquad w_i = \frac{g_i}{\sum_h g_h}. \tag{4}$$

Individual and group consistency use $CI = (\lambda_{\max} - 3)/2$ and $CR = CI/RI$, where $\lambda_{\max}$ is the largest eigenvalue, $RI = 0.52$, and the consistency criterion is $CR \leq 0.10$ (Saaty, 1994). All 17 complete responses contribute to the group weights, providing exploratory preference estimates.

Annual burdens $B_{mj} \in \{E_m, C_m, T_m\}$ are converted to linear scores and combined using these weights:

$$s_{mj} = 1 - \frac{B_{mj}}{U_j}, \qquad S_m = w_E s_{mE} + w_C s_{mC} + w_V s_{mV}, \tag{5}$$

with annual reference values $(U_E, U_C, U_V) = (24\ \text{kg}\,\text{CO}_2\text{e}, 360\ \text{SGD}, 1080\ \text{minutes})$. These scenario-defined references equal the assumed per-event values of 4 $\text{kg}\,\text{CO}_2\text{e}$, SGD 60, and 180 minutes multiplied by six events. They set the measurement scales, while AHP weights express criterion importance. A zero burden scores 1, and a burden equal to its reference scores 0; the linear functions extend across the burden domain. Higher scores indicate preferred outcomes.

2.4 Transport sensitivity analysis

Rental requires repeated collection and return, making travel a substantial component of its annual emissions, expenditure, and active time. The baseline assumes 24 dedicated one-way journeys, while collection and return may also be integrated with existing trips. Sensitivity analysis therefore examines the dependence of rental competitiveness on attributed travel burdens under fixed preference weights and scoring references.

Let $\alpha \in [0, 1]$ denote the share of baseline rental collection-and-return travel attributed to rental. Dedicated journeys correspond to $\alpha = 1$; fully shared journeys with zero incremental travel burden correspond to $\alpha = 0$. Travel emissions, fares, and active time change proportionally, while rental payments, cleaning, and handling remain fixed:

$$\begin{aligned} E_R(\alpha) &= 4.4916923 + 6.888\alpha, \\ C_R(\alpha) &= 60 + 48\alpha, \\ T_R(\alpha) &= 60 + 720\alpha. \end{aligned} \tag{6}$$

The constant terms comprise garment allocation and cleaning (4.4916923 kg CO_2e), six rental payments (SGD 60), and booking and handover (60 minutes). The travel coefficients follow from $240(0.0287) = 6.888$ kg CO_2e, $24(2) = 48$ SGD, and $24(30) = 720$ minutes. Ownership outcomes, criterion weights, and scoring references remain fixed. Rental–ownership crossover shares are obtained by equating their composite scores.

3 Results

3.1 Annual burdens and baseline ranking

Both purchase options yield 4.92 kg CO_2e/year, compared with 11.38 kg CO_2e/year for rental (Eq. 2; Figure 3). The common garment-stage allocation contributes 3.84 kg CO_2e/year. Purchase cleaning and delivery contribute 0.763 and 0.31844 kg CO_2e/year; rental cleaning and travel contribute 0.654 and 6.888 kg CO_2e/year. Annual expenditure is SGD 60.99 for second-hand, SGD 72.00 for new, and SGD 108.00 for rental (Table 2). Rental travel accounts for 60.5% of its emissions, 44.4% of its expenditure, and 92.3% of its active time, highlighting its influence across all three criteria.

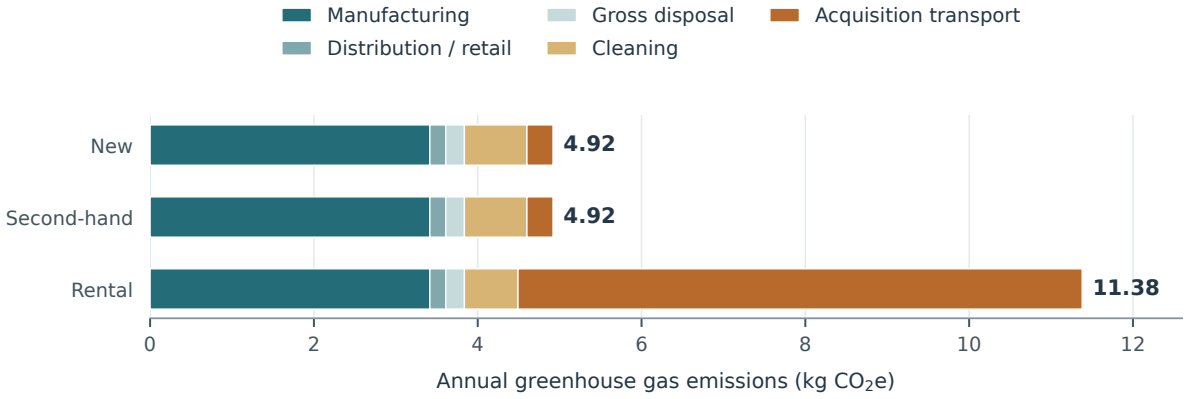


Figure 3: Annual greenhouse gas emissions by life-cycle stage. All channels use the common 26-wear garment allocation.

Table 2: Annual consumer expenditure by component (SGD).

Option	Garment	Delivery	Rent	Care	Travel	Total
New	65.00	0.00	0.00	7.00	0.00	72.00
Second-hand	52.00	1.99	0.00	7.00	0.00	60.99
Rental	0.00	0.00	60.00	0.00	48.00	108.00

The AHP weights are 0.2410 for emissions, 0.4295 for cost, and 0.3295 for convenience. Eleven of 17 individual matrices satisfy $CR \leq 0.10$, and the group matrix has $CR = 0.005885$. With these weights, second-hand purchasing scores 0.8412, followed by new purchasing at 0.8281 and rental at 0.5189 (Table 3).

Table 3: Baseline annual outcomes, criterion scores, and composite ranking.

Option	kg CO_2e/year	SGD/year	Minutes	s_E	s_C	s_V	S (rank)
Second-hand	4.92	60.99	120	0.7950	0.8306	0.8889	0.8412 (1)
New	4.92	72.00	120	0.7950	0.8000	0.8889	0.8281 (2)
Rental	11.38	108.00	780	0.5258	0.7000	0.2778	0.5189 (3)

Note. Minutes denote assumed active consumer time. Calculations retain full precision.

For second-hand purchasing, Eq. 5 gives criterion scores $1 - 4.9191323/24 \approx 0.7950$, $1 - 60.99/360 \approx 0.8306$, and $1 - 120/1080 \approx 0.8889$. Combining them gives

$$S_S \approx 0.2410(0.7950) + 0.4295(0.8306) + 0.3295(0.8889) \approx 0.8412.$$

Second-hand purchasing saves SGD 11.01 relative to new purchasing while sharing the same modelled emissions and active time. New purchasing has lower burdens than rental on all three criteria. These relationships establish the baseline ordering for positive criterion weights.

3.2 Transport and ranking changes

Substitution into Eq. 5 gives

$$S_R(\alpha) = 0.865006510 - 0.346091054\alpha. \tag{7}$$

Equating Eq. 7 to the fixed score S_m of each purchase option gives

$$\alpha_m^* = \frac{0.865006510 - S_m}{0.346091054}, \qquad m \in \{N, S\}. \tag{8}$$

Using full-precision scores, rental equals second-hand at $\alpha_S^* = 0.068704$ and new at $\alpha_N^* = 0.106662$ (Figure 4). Accordingly, rental ranks first below approximately 6.87% of baseline attributed travel burdens, second between approximately 6.87% and 10.67%, and third above approximately 10.67%. The intersecting alternatives share equal scores at each boundary.

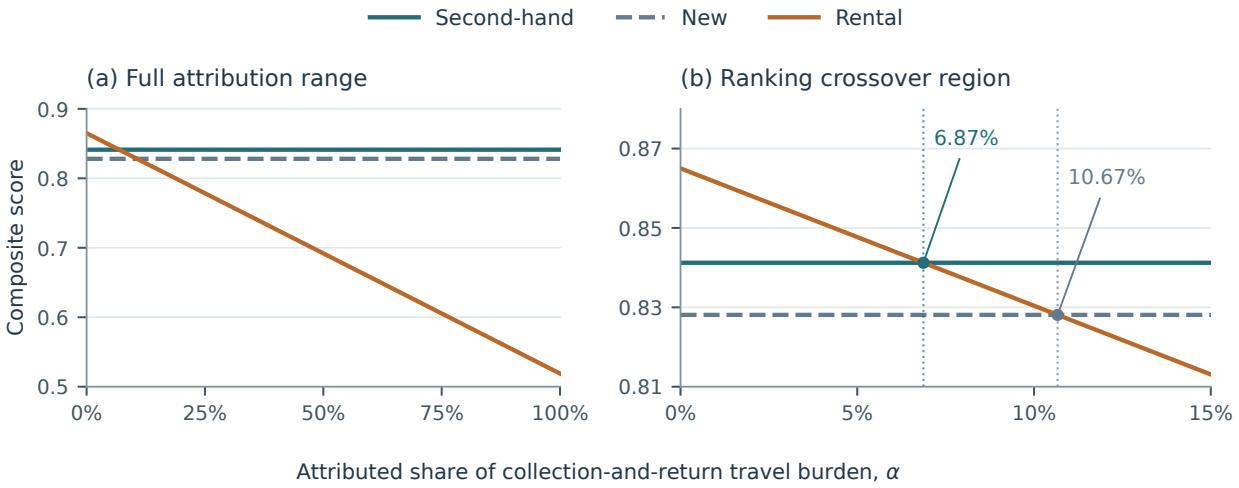


Figure 4: Composite scores as the attributed share of rental collection-and-return travel changes. Criterion weights and ownership outcomes remain fixed.

4 Discussion

The second-hand advantage reflects a lower acquisition price under common garment allocation, care, and delivery assumptions. Equal emissions for the purchase options follow from this allocation convention. Rental's repeated collection and return generate substantial travel costs and time, explaining its baseline position. Its first-place crossover near 7% identifies extensive travel sharing as a condition for competitiveness within the proportional scenario. Collection integrated with existing journeys therefore warrants attention in rental service design.

The estimates depend on the selected garment lifetime, advertised prices, proxy inventories, and assumed active time. The Italian care and UK transport factors provide screening estimates for the Singapore setting. Product-specific inventories, local routes, actual transaction costs, and channel-specific lifetimes would strengthen empirical precision. The original survey supports a small-sample assessment of criterion preferences. Individual CR describes within-respondent coherence, whereas group CR describes the aggregated matrix; the six individual matrices above the threshold remain represented in the exploratory group estimate. Transport thresholds also depend on the scoring references and the proportional relationship among travel emissions, fares, and time. In practice, trip sharing may affect these components by different proportions. The present analysis characterizes their joint response to a common attribution factor. These conditions define the interpretation of the modelled ranking.

5 Conclusion

For one student attending six formal events annually, second-hand purchasing achieves the highest AHP-weighted score, followed by new purchasing and rental. Its advantage arises from lower expenditure under equal modelled purchase-related emissions and time. Rental becomes the highest-scoring option when its attributed collection-and-return travel burdens fall below approximately 6.87% of the dedicated-travel baseline. The findings support second-hand acquisition under the specified assumptions and link rental competitiveness to efficient collection-and-return arrangements.

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Bibliographic note. Citation years follow the proceedings volume for Saaty (1994) and the journal volume for Monticelli and Costamagna (2023). Their online publication dates are June 1, 2009 and 2022, respectively. The source register provides the archived market and inventory evidence.

A Source and Reproduction Notes

A.1 Source locators and evidence status

Market sources are indexed as M2_001 (new), M2_002 (second-hand listing 1447640322), and M2_003 (rental listing 1464833759) in `data/sources.json`, with access dated 3 November 2026, Singapore time. Quotation extracts are archived in `evidence/market_quotes.md`. The baseline uses the first-rental refundable deposit of SGD 99.

Garment-stage emission factors refer to Sandin et al. (2019), printed page 83, Figure 4.24. The 0.478 kg reference mass appears on printed page 159, Table B-59. The cleaning proxy is the residential GWP entry in Monticelli & Costamagna (2023), Table 4, printed page 7635; Table 2 on page 7632 describes the associated inventory. Extracted figures and inventory records are retained in `evidence/environment/`, and `data/parameters.json` records each input's value, unit, and evidence category.

UK 2026 rail factors combine direct and upstream terms of $0.02121 + 0.00749 = 0.02870$ kg CO_2e/passenger-km from *Business travel–land* D89 and *WTT–pass vehs & travel–land* D86. Van factors combine $0.25716 + 0.06128 = 0.31844$ kg CO_2e/vehicle-km from *Delivery vehicles* D32 and *WTT–delivery vehs & freight* D31. The selected cells are archived in `evidence/environment/uk2026_selected_cells.json`.

A.2 Survey traceability and analytical records

The original workbook, `data/raw/survey_18.xlsx`, supplies Q8–Q10 from rows 2–19 and columns J–L of its response sheet. Cells B27–B29 in the question-and-rules sheet provide the wording transcribed in `results/run_manifest.json`. Seventeen records contain complete pairwise comparisons. Anonymous response identifiers, source rows, individual priorities, and CR values are retained in `results/ahp_individual.csv`; group priorities and consistency appear in `results/ahp_group.csv`. The annual outcomes, composite scores, and transport crossovers are summarized in `results/analysis_summary.json` and the accompanying CSV files. Reported values use full-precision calculations with rounding for presentation.

A.3 Data, reproducibility, and authorship

The repository retains the original questionnaire, parameter and source registers, respondent-level CR values, and full-precision results. Running `code/run.py` from the project root regenerates the numerical outputs; `report/generate_assets.py` recreates the result tables and generated figures while retaining the group-supplied LCA system-boundary PDF. Execution and compilation instructions accompany the repository. The source package contains the LaTeX manuscript and its referenced tables and figures, and the archived evidence records source locations and input hashes.

Author information. Yang Yihua, Cao Xiaoyue, Yang Liu, Liu Ziqing, and Xie Tongling. Faculty of Science, National University of Singapore. Instructor: Kim Leng Poh.

AI use statement. AI assistance supported English revision, LaTeX preparation, figure generation, and computational and document checks.